

Converting between the Three Forms of a Quadratic Expression (Homework)

Write the form of each quadratic expression below and convert to the other two forms.

$$\textcircled{1} y = 2(x+4)^2 - 2$$

$$\textcircled{2} -3(x-5)^2 + 27 = y$$

$$\textcircled{3} y = (x+3)^2 - 16$$

$$\textcircled{4} -(x-5)^2 + 36 = y$$

$$\textcircled{5} y = 4(x+4)(x+6)$$

$$\textcircled{6} -2(x-3)(x+7) = y$$

$$\textcircled{7} y = -(x+7)(x-5)$$

$$\textcircled{8} (x-2)(x-9) = y$$

$$\textcircled{9} y = 5x^2 + 40x + 35$$

$$\textcircled{10} y = -4x^2 + 48x - 80$$

$$\textcircled{11} x^2 + 4x - 32 = y$$

$$\textcircled{12} y = -6x^2 - 18x - 12$$

Answers

① Vertex

Standard: $y = 2x^2 + 16x + 30$

Intercept: $y = 2(x+5)(x+3)$

② Vertex

Standard: $y = -3x^2 + 30x - 48$

Intercept: $y = -3(x-2)(x-8)$

③ Vertex

Standard: $y = x^2 + 6x - 7$

Intercept: $y = (x-1)(x+7)$

④ Vertex

Standard: $y = -x^2 + 10x + 11$

Intercept: $y = -(x+1)(x-11)$

⑤ Intercept

Standard: $y = 4x^2 + 40x + 96$

Vertex: $y = 4(x+5)^2 - 4$

⑥ Intercept

Standard: $y = -2x^2 - 8x + 42$

Vertex: $y = -2(x+2)^2 + 50$

⑦ Intercept

Standard: $y = -x^2 - 2x + 35$

Vertex: $y = -(x+1)^2 + 36$

⑧ Intercept

Standard: $y = x^2 - 11x + 18$

Vertex: $y = (x - \frac{11}{2})^2 - \frac{49}{4}$

⑨ Standard

Intercept: $y = 5(x+1)(x+7)$

Vertex: $y = 5(x+4)^2 - 45$

⑩ Standard

Intercept: $y = -4(x-10)(x-2)$

Vertex: $y = -4(x-6)^2 + 64$

⑪ Standard, Intercept: $(x-4)(x+8) = y$

Vertex: $(x+2)^2 - 36 = y$

⑫ Standard, Intercept: $y = -6(x+1)(x+2)$

Vertex: $y = -6(x + \frac{3}{2})^2 + \frac{3}{2}$